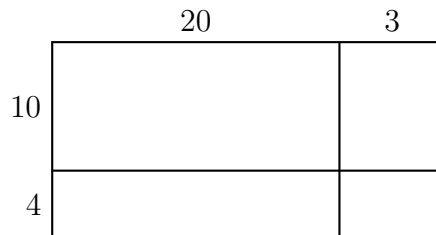


Area Models for Two-Digit Multiplication

Splitting factors, finding partial products, adding to the total, and estimating to check

Directions: An area model splits each factor into tens and ones, so one big multiplication becomes four easy cells. Labeled models are given in the first problems; later problems give you an empty model to label and fill in yourself.

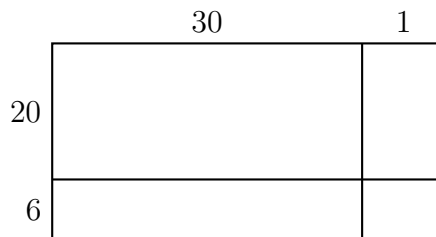
1. The model shows 14×23 , with 14 split into $10 + 4$ and 23 split into $20 + 3$. Find each cell, then add:



$$10 \times 20 = \underline{\hspace{2cm}} \quad 10 \times 3 = \underline{\hspace{2cm}} \quad 4 \times 20 = \underline{\hspace{2cm}} \quad 4 \times 3 = \underline{\hspace{2cm}}$$

Add the four cells: $14 \times 23 = \underline{\hspace{2cm}}$

2. The model shows 26×31 , split as $20 + 6$ and $30 + 1$:



$$20 \times 30 = \underline{\hspace{2cm}} \quad 20 \times 1 = \underline{\hspace{2cm}} \quad 6 \times 30 = \underline{\hspace{2cm}} \quad 6 \times 1 = \underline{\hspace{2cm}}$$

Add the four cells: $26 \times 31 = \underline{\hspace{2cm}}$

- 3.** Before you can build a model you must split the factors into tens and ones. Fill in the splits for 47×35 :

$$47 = 40 + \underline{\hspace{2cm}} \qquad 35 = 30 + \underline{\hspace{2cm}}$$

- 4.** Label the empty model for 24×15 (split each factor into tens and ones), fill in all four cells, and find the product.

$$24 \times 15 = \underline{\hspace{2cm}}$$

- 5.** Use the empty model for 32×28 , then list the four cells and their total:

$$\underline{\hspace{2cm}} + \underline{\hspace{2cm}} + \underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

6. First *estimate* 38×21 by rounding each factor to the nearest ten. Then use the model to find the exact product.

estimate: _____ exact: _____

7. This model is missing a side. The rows are 10 and 3, and two cells are already filled in:

	?
10	240
3	72

What is the unknown side length? _____

Check your answer using the bottom cell: $72 \div 3 =$ _____

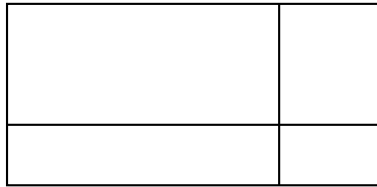
- 8.** A patio is 34 tiles long and 26 tiles wide. Use an area model to find how many tiles cover the whole patio.



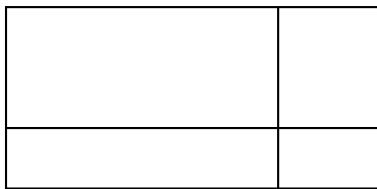
Answer: _____

- 9.** Elena computes 45×22 and writes 900. Use the model to find the correct product: _____

Then explain what Elena left out.



10. Challenge: use an area model to compute 63×47 .



$$60 \times 40 = \underline{\hspace{2cm}} \quad 60 \times 7 = \underline{\hspace{2cm}} \quad 3 \times 40 = \underline{\hspace{2cm}} \quad 3 \times 7 = \underline{\hspace{2cm}}$$

$$63 \times 47 = \underline{\hspace{2cm}}$$